Natural Language Processing EDS 6352
Homework 1 Report
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Case 1: Without lemmatization, removal of stop words, or handling logical negation (overall accuracy: 0.812)

Class	Precision	Recall	F1
Class 0	0.758	0.860	0.806
Class 1	0.838	0.725	0.777

Case 2: With lemmatization only (overall accuracy: 0.810)

Class	Precision	Recall	F1
Class 0	0.777	0.870	0.821
Class 1	0.852	0.750	0.798

Case 3: With lemmatization and removal of stop words (overall accuracy: 0.810)

Class	Precision	Recall	F1
Class 0	0.795	0.835	0.815
Class 1	0.826	0.785	0.805

Case 4: With lemmatization, removal of stop words, and handling logical negation (overall accuracy: 0.818)

Class	Precision	Recall	F1
Class 0	0.801	0.845	0.822
Class 1	0.836	0.790	0.812

1- Compare and discuss the per-class precision, recall, and F1 values for each of the 4 scenarios. Do you see any meaningful and significant improvement in these measures of accuracy for any scenario? What is your overall suggested preprocessing combination for sentiment analysis using Naïve Bayes?

I analyzed the processed text for all the four scenarios provided as it was the main difference provided to the Naive Bayes model. The overall accuracy score was similar for each scenario and not statistically significant.

One thing that is noticeable is that the precision is lower than the recall for Class 0 and precision is higher than the recall for Class 1 for all the scenarios. There isn't any significant class imbalance in the training data with 496 data points for class 0, and 504 data points for class 1. The possible issue that might be causing this is that the vocab size for the negative class might be a bit less. The results from the notebook do verify this claim with class 0 having 49020 words, while class 1 having 52507 words.

If all the scenarios are compared based on the F1 score, scenario 4 ranks the best among all. Overall it has a higher F1 score for both the classes: class 0 and class1.

2- Explain the effect of Laplace smoothing. What would happen if you had not used it?

I tried predicting the output from Naive Bayes without using Laplace smoothing, and I got a math domain error. Based on the theory, we compute likelihoods of words belonging to a class in Naive Bayes, but in the test data we might come across words that are not part of the training. If this happens, the likelihood for that word becomes zero. And since we are taking a log of the probabilities, taking a log of zero returns negative infinity.

3- Based on your results, identify and discuss limitations of Naïve Bayes classifier. This involves looking at the text of misclassified movie reviews to identify patterns of shortcomings of the classifier and/or its limitations.

The Naive Bayes classifier purely works on the word count, and calculating probabilities out of it. In the long reviews, it can become tricky for the classifier to pick up the sentiment. For eg. the main limitation I see in the misclassified example is that a reviews says good thing about the book, but not the movie, but the model predicts it as

positive. So to conclude when things such as sarcasm, current human trends, or slang gets included, it might get difficult for the classifier to identify those things.